GGG-S-00825A(GSA-FSS)
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SUPERSEDING
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INTERIM FEDERAL SPECIFICATION

SWAGING TOOL,

FLEXIBLE CABLE TERMINAL

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers one type of portable, pneumatic, swaging tool used to swage terminals to flexible aircraft cable.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-T-570 - Tool Steel, Alloy.
ZZ-H-500 - Hose, Rubber, and Hose Assemblies, Rubber; Pneumatic (Braided or Wrapped).
PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-1511 - Cable; Steel (Carbon) Flexible, Preformed.
MIL-C-4109 - Coupling Assembly, Air Hose, Quick Operating.
MIL-C-5424 - Cable; Steel (Corrosion-Resisting), Flexible, Preformed (for Aeronautical Use).
MIL-C-5693 - Cable; Steel (Corrosion-Resistant), Preformed (Aircraft Applications).
MIL-T-6117 - Terminal, Cable Assemblies, Swaged Type.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS20658 - Terminal, Wire Rope, Swaging, Fork End.
MS20663 - Ball End, Wire Rope, Swaging, Double Shank.
MS20664 - Ball End, Wire Rope, Swaging, Double Shank.
MS20667 - Terminal, Wire Rope, Swaging, Fork End.
MS20668 - Terminal, Wire Rope, Swaging, Eye End.
MS21259 - Terminal, Wire Rope, Stud.
MS21260 - Terminal, Wire Rope, Stud.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is descriptive, not restrictive, and is not intended to preclude the purchase of swaging tools otherwise conforming to this specification.

3.2 Preproduction model. As soon as practicable after award of the contract, the supplier shall furnish a swaging tool to prove, prior to starting production, that his production methods and choice of design detail will produce a swaging tool that complies with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the contracting agency shall not relieve the supplier of his obligation to furnish swaging tools conforming to this specification.

3.3 Standard product. Each swaging tool furnished under this specification shall be a currently standard product of an established manufacturer, except for any deviations from the manufacturer's standard product that are required by this specification. All accessories and components normally furnished commercially with the standard product offered under this specification shall be in the same quantity and of the same quality as furnished commercially with the standard product.

3.4 Material. Materials not specified shall be of a quality suitable for the intended purpose. All materials shall be free from defects and imperfections that would affect the serviceability of the finished product.

3.5 Design and construction. The swaging tool shall be designed for portable, hand operation and shall be similar to that shown in figure 1. The tool shall consist essentially of a cylinder and yoke assembly, hose assembly, foot control valve assembly, and the swaging and cutting dies. The tool shall operate on an air pressure of 80 to 100 PSI and shall be capable of completing all the tests specified herein. The complete tool, minus the dies, shall weigh not more than 20 pounds. The tool shall be furnished with the gages and the instruction manual in a metal carrying case.

3.5.1 Cylinder and yoke assembly. The cylinder and yoke assembly shall be cylindrical in shape and mounted on a cover and cushion to facilitate bench mounting. The exposed surfaces of the cylinder and yoke shall be plated in accordance with QQ-C-320. The yoke shall hold the dies in position while swaging and cutting and be constructed to permit rapid changing of the dies without the use of tools. The yoke shall be equipped with a guard (not shown in figure 1) which covers the die opening and protects the operator from accidental injury. The guard shall not interfere with the swaging or cutting operation.

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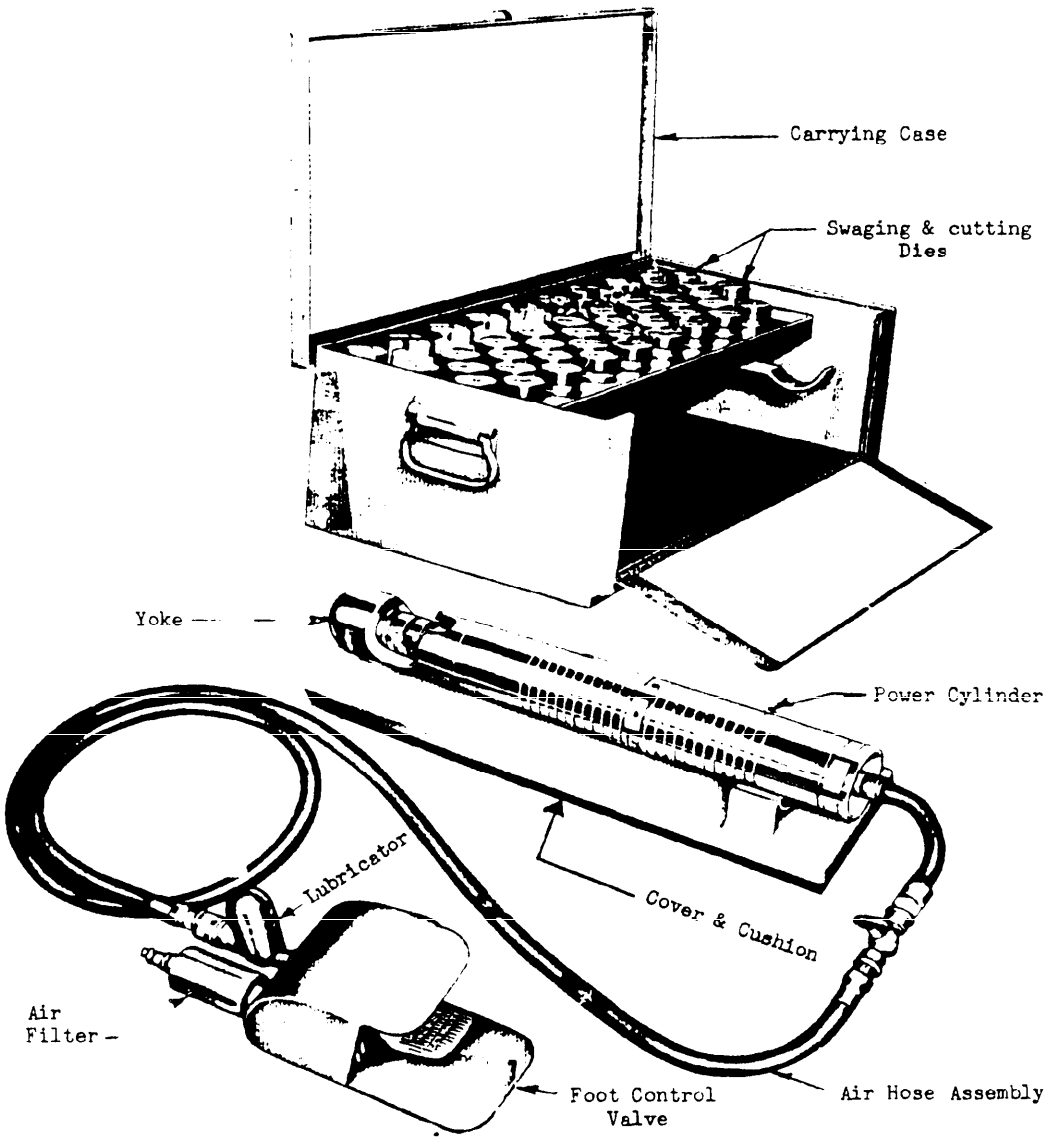


FIGURE 1. Swaging tool, dies and case.

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3.5.2 Air Hose Assembly. The air hose assembly shall be approximately 5 feet in overall length and shall include a shut-off valve. The hose shall conform to ZZ-H-500, Class 2, 3/8-inch inside diameter. One end of the hose shall be fitted with a 1/4-inch threaded pipe fitting for securing hose to the cylinder and the other end fitted with a quick operating coupling conforming to MIL-C-4109 for attaching hose to foot control valve.

3.5.3 Foot control valve assembly. The foot control valve shall be the single action, on-off type for control of the supply of air to the cylinder. The valve shall be enclosed in a heavy duty steel frame designed to prevent inadvertent activation of the valve. The valve assembly shall include a lubricator and an air filter similar to those shown in figure 1. The lubricator shall be the automatic feed type capable of supplying lubricant to the cylinder and yoke and have a reservoir of sufficient capacity so that one filling will last for a minimum of eight hours operation of the swaging tool. The air filter, located at the air inlet of the foot control valve, shall be capable of retaining solid particles, 80 microns and larger in size, and shall contain a removable, reusable filter cartridge. All exposed metal parts of this assembly shall be plated to prevent corrosion in accordance with QQ-P-416.

3.5.4 Swaging and cutting dies. The swaging and cutting dies shall be furnished as matched pairs and designed so that they can be readily and quickly installed in and removed from the yoke without the use of tools. The swaging dies shall be made of Class S5 tool steel and the cutting dies made of Class S2 tool steel conforming to QQ-T-570 and shall be heat treated to resist high impacting and hardened to Rockwell "C" 57 to 60. All functional surfaces of the dies shall have a 10 microinch, maximum surface roughness and shall be highly polished. They shall be marked in a legible and permanent manner to indicate the type and size of terminal to be swaged or the size cable to be cut, as applicable, and to insure their use in matched pairs. The dies shall be capable of completing the performance tests specified herein.

3.5.4.1 Swaging dies. Twenty pairs of swaging dies shall be furnished the type and size as follows: one pair for each type of terminal shown on MS20663 and MS20664 in the 5 cable sizes 1/16 thru 3/16 inch (10 pairs); one pair for all 5 types of terminals shown on MS20260, MS20658, MS20667, MS20668, and MS21259 in the 7 cable sizes 1/16 thru 1/4 inch (7 pairs); and one pair for a commercial plain ball type terminal in the cable sizes of 1/16, 3/32 and 1/8 inch (3 pairs).

3.5.4.2 Cutting dies. Seven pairs of cutting dies shall be furnished, one pair for each of the following cable diameter sizes 1/16, 3/32, 1/8, 5/32, 3/16, 7/16 and 1/4 inch.

3.5.5 Gages. The gages shall be furnished as a set consisting of a minimum of 17 gages contained in a metal holder. The gages shall be flat and straight and the "Go" type for use in determining the acceptability of swaged terminals. They shall be made of tool steel, hardened to Rockwell "C" 60 to 62. The dimension of the gage opening shall conform to the basic "after swaging" dimension shown on the applicable MS drawing with a tolerance of plus .0000 minus .0002 inch. Each gage shall be marked in a legible and permanent manner to indicate the type and size terminal and the dimension to be gaged as follows: on one side of the gage the marking shall include the applicable MS part number of the terminal and the words "ball only" or "shank only" as applicable; the other side of the gage shall have the same marking except the AN part number shall be used in lieu of the MS part number. (i.e., AN663-2, AN664-2 SHANK ONLY and MS20663-2, MS20664-2 SHANK ONLY) The opening dimension of the gages for the plain ball type terminals shall conform to the "spherical diameter" dimension shown on MS20663 and the gage shall be marked with the same commercial designation as shown in the instruction manual.

3.5.6 Carrying Case. The carrying case shall be made of 20 gage, minimum, sheet steel and shall be similar to that shown in figure 1. The case shall be large enough to contain the complete swaging tool and other specified equipment and shall be painted in accordance with the manufacturer's standard practice. A separate compartment for each die shall be provided to retain the die when case is in transit. The case shall have a carrying handle on each end and latch to hold the case lid in a closed position. The case shall have an identification plate secured to the inside of the lid. The plate shall contain the manufacturer's name, address and model or serial number of the swaging tool.

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3.6 Performance.

3.6.1 Swaging. The swaging tool when equipped with the appropriate swaging dies shall be capable of swaging the terminals specified in 3.5.4.1 to the cables specified below. The swaging is accomplished by uniformally cold forming the terminal to the cable. The swaged terminal shall conform to the basic "after swaging" dimensions and the minimum breaking strength shown on the applicable MS drawing.

3.6.2 Cutting. When equipped with the appropriate cutting dies, the tool shall be capable of cutting cable conforming to MIL-C-1511, MIL-C-5424 and MIL-C-5693 in the sizes specified in 3.5.4.2. The tool shall cut the cable without distorting, unraveling or otherwise make additional operations necessary prior to inserting the cable in the terminal.

3.7 Instruction book. One copy of an instruction book shall be furnished with each swaging tool. The book shall contain complete operating instructions and parts list. The data for the swaging dies shall reference the appropriate AN and the MS part numbers or the commercial designation as applicable.

3.8 Workmanship. The swaging tool shall be constructed and assembled in a thoroughly workmanlike manner to assure good quality equipment and general appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection:

(a) Preproduction inspection. Preproduction inspection shall consist of the examinations specified in 4.3 and all the tests specified in 4.4.

(b) Acceptance inspection. Acceptance inspection shall consist of the examinations specified in 4.3.

4.3 Examinations.

4.3.1 Visual examination. The complete swaging tool including all specified equipment shall be examined for conformance in design and construction, material, finish, workmanship, and marking. (See 3.4, 3.5 thru 3.5.6, 3.7 and 3.8) Presence of one or more of the following defects shall be cause for rejection:

1. Component part missing.
2. Material not as specified.
3. Quantity of dies not as specified.
4. Markings on dies, gages, or case not as specified.
5. Paint or plating not as specified.
6. Surface roughness or polished surfaces not as specified.
7. Excessive weight.
8. Dimensions not as specified.

4.4 Tests.

4.4.1 Cutting test. The swaging tool shall be assembled following the manufacturer's procedure and connected to an air supply of 90 PSI. One pair of cutting dies shall be installed in the yoke and 100 cuts shall be made using the appropriate size cable conforming to MIL-C-1511. The last 5 pieces of cut cable shall be examined and inserted into a sample MS terminal to determine conformance to the requirements of 3.6.2. Any nonconformance to the stated requirements shall be cause for rejecting the tool. Each pair of cutting dies shall be subjected to the same kind of test, examination, and acceptance.

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4.4.2 Swaging test. The swaging test shall be conducted with the tool assembled as described in 4.4.1 and with a pair of swaging dies installed in the yoke. The terminal, with the cable inserted, shall be placed between the dies and while swaging, rotated 180 degrees, axially, to assure uniform swaging. The swaging time shall not exceed the time specified in table 1 by more than 10 percent. Swaged terminals shall have a smooth surface free from die marks and rapid dimensional changes in cross-section, and shall be free from splits, cracks, or other defects. The swaged terminal shall be measured with the appropriate gage or gages as applicable. Any terminal failing to pass thru the gage, with a minimum effort, or shows any evidence of the above mentioned defects shall be classified as defective and be cause for rejecting the swaging tool.

TABLE I. Terminal sizes and swaging time

Terminal	Dash No's. 1/	Nom. cable size (inches)	Swaging time (sec.)
MS 20658 MS 20667 MS 20668 MS 21259 MS 21260	-2 -3 -4 -5 -6 -7 -8	1/16 3/32 1/8 5/32 3/16 7/32 1/4	15 15 20 35 45 60 100
MS 20663 MS 20664	-2 -3 -4 -5 -6	1/16 3/32 1/8 5/32 3/16	6 8 10 30 100
Plain Ball	- - -	1/16 3/32 1/8	5 6 8

1/ - Denotes dash numbers included in each MS.

4.4.2.1 The same test shall be conducted on 5 terminals using each of the 20 pairs of dies specified in 3.5.4.1 and using each type of cable specified in 3.6.2. Each swaged terminal shall be gaged and classified as stated above. If all gaged terminals are acceptable, they shall then be subjected to the pull test specified in 4.4.2.2.

4.4.2.2 Each swaged terminal-cable assembly shall be subjected to the following pull test. The terminal shall be secured to a stable fixture and a load, equal to the minimum breaking strength specified on the applicable MS drawing, shall be applied to the cable. Prior to applying the load, the cable shall be marked at the point where it enters the swaged terminal. While the load is applied, the cable marking shall be examined to determine if any slippage has occurred. Any evidence of slippage between the cable and terminal on any sample being tested shall be cause for rejecting the swaging tool.

4.4.2.3 Upon completion of the cutting and swaging tests, the swaging tool shall be disassembled and the various parts examined. Any evidence of the following shall be cause for rejecting the swaging tool:

1. Malfunction of component part.
2. Improper lubrication.
3. Nicked, cracked, or deformed component.
4. Evidence of wear on dies.
5. Pitted or scratched areas on the polished surface of swaging dies.
6. Leaking of oil or air.

4.5 Inspection of preparation for delivery. The inspections for preparation for delivery shall be in accordance with PPP-P-40.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Swaging tools shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. The portable swaging tool covered by this specification is intended to be used for cutting aircraft cable and for swaging terminals to aircraft cable.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Applicable levels of preservation, packaging, and packing (see 5.1).

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to: General Services Administration, FSS (FMSK), Wash., DC 20406.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

GGG-S-00825A(GSA-FSS) Swaging Tool, Flexible Cable Terminal

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM GSA FEB 69 2200

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